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VEHICLE CONTROL DEVICE AND VEHICLE CONTROL METHOD

Abstract

A vehicle control device capable of autonomously controlling the steering of a vehicle includes: a necessity determination unit that determines whether offset travel offset from a center of a traveling lane to a side opposite to an opposite lane side is necessary based on a state of an oncoming lane; an execution determination unit that determines whether an execution condition for executing offset travel is satisfied based on a road state on a side opposite to an oncoming lane side of the vehicle when offset travel is necessary; and a vehicle control unit that controls the vehicle. When it is determined that the execution condition is satisfied, the vehicle control unit causes the vehicle to travel by the offset travel, and when it is determined that the execution condition is not satisfied, the vehicle control unit performs at least one of restricting the autonomous driving function and notifying the occupant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032982 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a vehicle control device and a vehicle control method.

2. Description of Related Art

[0003] Conventionally, there is known a vehicle capable of autonomous steering, the vehicle being caused to travel while being shifted from the center of a lane in which the vehicle is traveling within the width of the lane (Japanese Unexamined Patent Application Publication No. 2019-123402, Japanese Unexamined Patent Application Publication No. 2021-099541, etc.). In the device described in JP 2019-123402 A, in particular, when there is another vehicle traveling in an adjacent lane adjacent to the lane in which the vehicle is traveling, the target travel trajectory of the vehicle is corrected so as to be laterally away from the other vehicle. The correction of the target travel trajectory is performed based on information on the environment on the side opposite to the adjacent lane.

SUMMARY

[0004] On a road with two-way traffic, particularly when a median strip is not provided, an oncoming vehicle travels near the host vehicle in the lateral direction. In such a case, the occupant may feel anxious if autonomous steering control for autonomous steering is performed. On the other hand, when the vehicle is caused to travel in an offset manner away from an opposite lane on a road with two-way traffic, the distance from a lane marking or any obstacle may become too short on the side of the lane in which the host vehicle is traveling opposite to the opposite lane. This may make the occupant feel anxious.

[0005] In view of the above issue, an object of the present disclosure is to suppress an occupant feeling anxious when the host vehicle approaches an oncoming vehicle or the like during autonomous steering control for autonomous steering.

[0006] The gist of the present disclosure is as follows.

[0007] (1) A vehicle control device that is able to autonomously control steering of a vehicle, the vehicle control device including: [0008] a necessity determination unit that determines necessity of offset travel in which the vehicle is offset from a center of a traveling lane, in which the vehicle is traveling, to a side opposite to an opposite lane, in which a vehicle travels in a direction opposite to the traveling lane, based on a situation of the opposite lane; [0009] an execution determination unit that determines whether an execution condition for executing the offset travel is met based on a road condition on a side of the vehicle opposite to the opposite lane when the offset travel is necessary; and [0010] a vehicle control unit that controls the vehicle, in which: [0011] the vehicle control unit causes the vehicle to travel using the offset travel when it is determined that the execution condition is met when the offset travel is necessary; and [0012] the vehicle control unit performs at least one of restricting an autonomous driving function and notifying an occupant when it is determined that the execution condition is not met when the offset travel is necessary.

[0013] (2) A vehicle control method that is able to autonomously control steering of a vehicle, the vehicle control method including: [0014] determining necessity of offset travel in which the vehicle

is offset from a center of a traveling lane, in which the vehicle is traveling, to a side opposite to an opposite lane, with traffic in a direction opposite to the traveling lane, based on a situation of the opposite lane; determining whether an execution condition for executing the offset travel is met based on a road condition on a side of the vehicle opposite to the opposite lane when the offset travel is necessary; [0015] causing the vehicle to travel using the offset travel when it is determined that the execution condition is met when the offset travel is necessary; and [0016] performing at least one of restricting an autonomous driving function and notifying an occupant when it is determined that the execution condition is not met when the offset travel is necessary. [0017] According to the present disclosure, it is possible to suppress an occupant feeling anxious when the host vehicle approaches an oncoming vehicle or the like during autonomous steering control for autonomous steering.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0019] FIG. 1 is a schematic block diagram illustrating a configuration of a vehicle control system;

[0020] FIG. 2 is a functional diagram of a processor of an ECU;

[0021] FIG. 3 is a diagram schematically illustrating a state in which a vehicle is traveling in a traveling lane;

[0022] FIG. 4 is a view similar to FIG. 3 showing the vehicle traveling in a traveling lane;

[0023] FIG. 5 is a flow chart of a steering process for vehicles; and

[0024] FIG. 6 is a flowchart related to a steering process of a vehicle according to a first modification.

DETAILED DESCRIPTION OF EMBODIMENTS

[0025] Hereinafter, embodiments will be described in detail with reference to the drawings. In the following description, the same reference numerals are given to the same constituent elements.

Configuration of Vehicle Control System

[0026] Referring to FIG. 1, a configuration of a vehicle control system 1 including a vehicle control device according to an embodiment will be described. The vehicle control device is a device that controls the vehicle 100, and in particular, can autonomously control acceleration/deceleration and steering of the vehicle 100. FIG. 1 is a schematic block diagram illustrating a configuration of a vehicle control system 1.

[0027] The vehicle control system 1 is mounted on the vehicle 100 and controls acceleration/deceleration and steering of the vehicle 100. As illustrated in FIG. 1, in the present embodiment, the vehicle control system 1 includes an outside camera 11, a ranging sensor 12, a positioning sensor 13, a vehicle sensor 14, a human machine interface (HMI) 15, a vehicle actuator 21, and an electronic control unit (hereinafter, referred to as “ECU”) 30. The outside camera 11, the ranging sensor 12, the positioning sensor 13, the vehicle sensor 14, HMI 15, and ECU 30 are communicably connected via, for example, the in-vehicle network 25. The in-vehicle network 25 is, for example, a network conforming to a standard such as CAN (Controller Area Network). Further, ECU 30 is connected to the vehicle actuator 21 via a signaling line.

[0028] The outside camera 11 is a device that captures an image of the surroundings of the vehicle. In the present embodiment, the outside camera 11 captures an image of the front of the vehicle 100. The outside camera 11 is, for example, a CMOS (Complementary Metal-Oxide-Semiconductor) camera or a CCD (Charge-Coupled Device) camera that is sensitive to visible-light. In the present embodiment, the outside camera 11 is mounted in the vehicle of the vehicle 100, for example, so as

to face the front of the vehicle **100**. The outside camera **11** captures an image of a front region of the vehicle **100** at predetermined imaging cycles, and generates an image in which the front region is captured. Each time an image is generated, the outside camera **11** outputs the generated image to ECU **30** via the in-vehicle network **25**.

[0029] The ranging sensor **12** is a sensor that measures a distance to an object existing around the vehicle **100**. In the present embodiment, the ranging sensor **12** measures a distance to an object existing in front of the vehicle **100**. For example, the ranging sensor **12** may measure a distance to a preceding vehicle of the vehicle **100**. The ranging sensor **12** is, for example, a radar such as a millimeter-wave radar or a LIDAR. The ranging sensor **12** outputs distance-information to the surrounding objects at predetermined intervals to ECU **30** via the in-vehicle network **25**.

[0030] The positioning sensor **13** is a device that measures the self-position of the vehicle **100**. The positioning sensor **13** is, for example, a GNSS receiver. GNSS receiver receives GNSS signals from the plurality of GNSS satellites and measures the self-position of the vehicle **100** based on the received GNSS signals. The positioning sensor **13** outputs the self-position information of the vehicle **100** to ECU **30** via the in-vehicle network **25** at predetermined intervals. Note that the positioning sensor **13** may be a receiver conforming to another satellite positioning system as long as it can measure the self-position of the vehicle **100**.

[0031] The vehicle sensor **14** is a sensor that detects a state of the vehicle **100**. The vehicle sensor **14** detects a traveling state and an operation state of the vehicle **100**. The vehicle sensor **14** includes, for example, a speed sensor, an acceleration sensor, and a yaw rate sensor as sensors for detecting the traveling state of the vehicle **100**. The speed sensor detects the speed of the vehicle **100**. The acceleration sensor detects acceleration of the vehicle **100**. The yaw rate sensor detects a change speed (yaw rate) of the yaw angle when the vehicle **100** turns. In addition, the vehicle sensor **14** includes, for example, a grip sensor that detects whether the driver grips the steering wheel (or touches the steering wheel) as a sensor that detects an operation state of the vehicle **100**. The vehicle sensor **14** outputs the detection result to ECU **30** at predetermined intervals via the in-vehicle network **25**.

[0032] HMI **15** is an interface for inputting and outputting data between a driver or an occupant and the vehicle control system **1**. HMI **15** includes an information providing device for providing various types of information to a driver or an occupant, and an input device for the driver or the occupant to perform an input operation.

[0033] Specifically, HMI **15** includes a display **16** for displaying character information or picture information as an information providing device. The display **16** is any type of display device such as a liquid crystal display or an organic EL display. The display **16** receives an image signal from ECU **30** via the in-vehicle network **25**, and displays an image in accordance with the image signal. HMI **15** also includes a speaker **17** for outputting sound as an information-providing device. The speaker **17** receives an audio signal from ECU **30** via the in-vehicle network **25**, and outputs audio in accordance with the audio signal. HMI **15** may include, as the information providing device, a device (for example, a vibrating device) other than the display **16** and the speaker **17** that provides various kinds of information to the drivers or the occupants.

[0034] In addition, HMI **15** has, as an input device, a touch panel **18** on which input is performed by a driver or a passenger touching. The touch panel **18** outputs an operation signal to ECU **30** via the in-vehicle network **25** when the driver or the occupant performs an operation by touching. HMI **15** may include, as the input device, a device (for example, a button, a switch, or the like) other than the touch panel **18** for the driver or the occupant to perform the input manipulation.

[0035] The vehicle actuator **21** is an actuator used to control the operation of the vehicle **100**. Specifically, the vehicle actuator **21** includes, for example, a drive actuator, a braking actuator, and a steering actuator. The drive actuator controls an internal combustion engine or an electric motor for driving the vehicle **100**. The braking actuator controls a brake for braking the vehicle **100**. The steering actuator controls steering of the vehicle **100**. The vehicle actuator **21** controls acceleration,

braking, and steering of the vehicle **100** in accordance with a control signal transmitted from ECU **30** via a signal line.

Configuration of Vehicle Control Device

[0036] ECU **30** functions as a vehicle control device that autonomously controls acceleration/deceleration and steering of the vehicle **100**. Thus, ECU **30** controls the actuation of the vehicle actuator **21**. In particular, in the present embodiment, ECU **30** autonomously controls the steering of the vehicle **100** (autonomous steering control) when there is an autonomous steering demand for autonomously controlling the steering from the user. At this time, for example, ECU **30** controls the steering of the vehicle **100** so that the vehicle **100** travels along the center in the width direction of the traveling lane in which the vehicle **100** is traveling (hereinafter, the center in the width direction of the lane is also simply referred to as the “center of the lane”).

[0037] Further, ECU **30** controls information to be provided from the information providing device of HMI **15**. Therefore, ECU **30** controls the images displayed on the display **16** and the sound outputted from the speaker **17**. As illustrated in FIG. 1, ECU **30** includes a communication interface **31**, a storage unit **32**, and a processor **33**.

[0038] The communication interface **31** is a circuit for connecting ECU **30** to the in-vehicle network **25**. The storage unit **32** stores data. The storage unit **32** includes, for example, at least one of a volatile semiconductor memory, a nonvolatile semiconductor memory, a hard disk drive (HDD), and a solid state drive (SSD). The storage unit **32** stores a computer program executed by the processor **33** of ECU **30**. Further, the storage unit **32** stores data used in a computer program executed by the processor **33**, such as data transmitted from the outside camera **11** or the like.

[0039] The processor **33** comprises one or more CPU (Central Processing Unit) and its peripheral circuitry. The processor **33** may further include other arithmetic circuits such as a logical arithmetic unit or a numerical arithmetic unit. The processor **33** executes a computer program stored in the storage unit **32**.

[0040] FIG. 2 is a functional diagram of the processor **33** of ECU **30**. As illustrated in FIG. 2, the processor **33** includes a necessity determination unit **331**, an execution determination unit **332**, and a vehicle control unit **333**.

[0041] The necessity determination unit **331** determines whether or not offset travel is necessary, which is offset from the center of the traveling lane to the opposite side of the traveling lane side, during execution of the autonomous steering control, on the basis of the state of the opposite lane on which the vehicle travels in a direction opposite to the traveling lane on which the vehicle **100** is traveling. In particular, in the present embodiment, the necessity determination unit **331** determines that offset travel is necessary when the distance in the width direction between the traveling lane and the oncoming lane is short or when the distance in the width direction between the vehicle **100** and the oncoming vehicle traveling on the oncoming lane is short. The front image generated by the outside camera **11** is input to the necessity determination unit **331**. Further, the self-position information measured by the positioning sensor **13** may be input to the necessity determination unit **331**. On the other hand, the necessity determination unit **331** outputs the necessity of the offset travel to the execution determination unit **332**.

[0042] In the present embodiment, the state of the oncoming lane includes a distance D between the center of the traveling lane and the end of the oncoming lane on the traveling lane side in the width direction (hereinafter, referred to as “distance from the end of the oncoming lane”). Therefore, the necessity determination unit **331** determines the necessity of the offset travel based on the distance D from the end of the oncoming lane. In particular, in the present embodiment, the necessity determination unit **331** determines that offset travel is required when the distance D from the end of the oncoming lane is less than the predetermined reference distance Dref. On the other hand, when the distance D from the end of the oncoming lane is equal to or larger than the predetermined reference distance Dref, the necessity determination unit **331** determines that the offset-traveling is not necessary.

[0043] FIG. 3 is a diagram schematically illustrating a state in which the vehicle **100** is traveling in the traveling lane Ld. The road shown in FIG. 3 is a face-to-face road, and the opposite lane Lo is located next to the traveling lane Ld in the widthwise direction.

[0044] In the embodiment illustrated by a broken line in FIG. 3, the distance D between the center Cld of the traveling lane Ld and the end of the opposite lane Lo (the distance between the end of the opposite lane) is D1. In FIG. 3, it is assumed that the vehicle **100** travels in the center in the widthwise direction in the traveling lane Ld during the autonomous steering control. However, in the embodiment shown in FIG. 3, the distance D1 is a distance that is shorter than the reference distance Dref. Therefore, the necessity determination unit **331** determines that the vehicle **100** needs to travel offset from the center of the traveling lane Ld in the vehicle width direction toward an opposite side from the oncoming lane Lo side (away from the oncoming lane side).

[0045] Specifically, the necessity determination unit **331** performs image processing on the front image input from the outside camera **11** to recognize the division line. An arbitrary recognition method such as a Hough transform is used to recognize the division line. In addition, the necessity determination unit **331** recognizes the traveling lane and the oncoming lane on the basis of the recognized lane division. Determination of whether each lane sandwiched between the two lanes is a traveling lane or an opposite lane is performed based on a line type (a difference in a solid line, a broken line, a color, and the like) of the lane, the number of lanes, and the like obtained by the recognition processing of the lane. In addition, in a case where a high-precision map including the number of lanes, the position, and the like is stored in the storage unit **32**, the traveling lane and the oncoming lane may be recognized on the basis of the position information detected by the high-precision map and the positioning sensor **13** in addition to the recognized division line.

[0046] Thereafter, the necessity determination unit **331** determines whether the traveling lane Ld of the vehicle **100** is a lane adjoining the oncoming lane Lo. Then, when the traveling lane Ld of the vehicle **100** is a lane adjoining the oncoming lane Lo, the necessity determination unit **331** calculates a distance D between the division line of the oncoming lane Lo on the traveling lane Ld side and the center of the traveling lane Ld. The lane division of the oncoming lane Lo on the traveling lane Ld side is recognized by the recognition process described above, and the center of the traveling lane Ld is calculated based on the lane divisions that are on both sides of the vehicle **100** and recognized by the recognition process. Then, the necessity determination unit **331** determines whether the distance D calculated in this way is less than a preset reference distance Dref, and determines that offset-running is required when the distance D is less than the reference distance Dref.

[0047] In the present embodiment, as the condition of the oncoming lane, the necessity of the offset travel is determined by using the distance (distance from the end of the oncoming lane) D between the center of the traveling lane Ld and the end of the oncoming lane Lo on the traveling lane Ld in the widthwise direction. However, as long as the distance in the width direction between the traveling lane Ld and the oncoming lane Lo and the distance in the width direction or the distance between the vehicle **100** and the oncoming vehicle traveling on the oncoming lane Lo is expressed, a parameter other than the distance D to the end of the oncoming lane may be used as the condition of the oncoming lane.

[0048] Specifically, as the condition of the oncoming lane, a widthwise distance between the center of the traveling lane Ld and the center of the oncoming lane Lo may be used. In addition, as the condition of the oncoming lane, a distance in the width direction between an end of the traveling lane Ld on the oncoming lane Lo side in the width direction and an end of the oncoming lane Lo on the traveling lane Ld side in the width direction may be used. Alternatively, as the condition of the opposite lane, a distance in the width direction between the end of the traveling lane Ld on the opposite lane Lo side and the center of the opposite lane Lo may be used as the condition of the opposite lane.

[0049] In addition, as the condition of the oncoming lane, the distance between the vehicle **100** (the

end on the center or on the oncoming lane Lo) and the oncoming vehicle X (the end on the center or on the traveling lane Ld) traveling in the oncoming lane Lo may be used. Alternatively, as the condition of the oncoming lane, the distance between the vehicle **100** (the center or the oncoming lane Lo side end) and the oncoming lane Lo (the center or the traveling lane Ld side end) may be used. Further, as the condition of the oncoming lane, the distance between the traveling lane Ld (the center or the oncoming lane Lo side end) and the oncoming vehicle X (the center or the traveling lane Ld side end) may be used. Even when any distance is used as the condition of the oncoming lane, it is determined that offset travel is necessary when the distance is less than the reference distance, and it is determined that offset travel is not necessary when the distance is equal to or greater than the reference distance.

[0050] In addition, the presence or absence of the oncoming vehicle X may be used as a situation of the oncoming lane. In this case, for example, when the oncoming vehicle X is traveling, the reference distance is set to be longer than when the oncoming vehicle X is not traveling.

[0051] When the offset travel is required, the execution determination unit **332** determines whether or not the execution condition for executing the offset travel is satisfied based on the road condition on the other side of the vehicle **100** from the oncoming lane Lo. In the present embodiment, the execution determination unit **332** determines that the execution condition is satisfied when the distance between the vehicle **100** and the demarcation line is apart from the oncoming lane Lo of the vehicle **100**, when the distance between the vehicle **100** and the curbstone is short, or when the distance between the vehicle and an obstacle located on the road shoulder or the like is short. The execution determination unit **332** receives the front image generated by the outside camera **11** and the distance information of the ranging sensor **12**. Further, the self-position information measured by the positioning sensor **13** may be input to the execution determination unit **332**. On the other hand, the execution determination unit **332** outputs the determination result of the execution condition success/failure of the offset travel to the vehicle control unit **333**.

[0052] Referring to FIG. **4**, the roadway condition on the other side of the vehicle **100** from the oncoming lane Lo side will be described. FIG. **4** is a view similar to FIG. **3**, showing a state in which the vehicle **100** is traveling in the traveling lane Ld. The road shown in FIG. **4** is also a face-to-face road, and the opposite lane Lo is located next to the traveling lane Ld in the widthwise direction.

[0053] In the present embodiment, the road condition on the opposite side of the vehicle **100** from the opposite lane Lo side includes distance $d1$ between the dividing line on the opposite side of the vehicle **100** from the opposite lane Lo side and the vehicle **100** when it is assumed that the vehicle **100** travels at a position in the reference distance D_{ref} from the end of the opposite lane Lo (hereinafter referred to as “reference position traveling”). If this range $d1$ is too close, the vehicle **100** is more likely to touch other vehicles traveling in neighboring lanes or obstacles located outside of this demarcation line. Therefore, the distance $d1$ is equal to or larger than the predetermined lane reference distance d_{ref1} .

[0054] In the present embodiment, the road condition on the side opposite to the oncoming lane Lo side of the vehicle **100** includes the distance $d2$ between the curbstone (or another step defining a boundary with a sidewalk or the like) Y located on the side opposite to the oncoming lane Lo side of the vehicle **100** and the vehicle **100** while traveling at the reference position. If the distance $d2$ is too short, the possibility of the vehicle **100** contacting the step such as the curbstone Y is increased, and therefore, the distance $d2$ is equal to or larger than the predetermined step reference distance d_{ref2} .

[0055] Further, in the present embodiment, the road condition on the side opposite to the oncoming lane Lo side of the vehicle **100** includes a distance $d3$ between an obstacle (a pedestrian, a bicycle, a roadside object, a stopped vehicle, or the like) Z located on the side opposite to the oncoming lane Lo side of the vehicle **100** and the vehicle **100** while traveling at the reference position. When the distance $d3$ is too short, the possibility that the vehicle **100** touch the obstacle Z is increased,

and therefore, the distance **d3** is equal to or larger than the predetermined obstacle reference distance **dref3**. The obstacle reference range **dref3** may be changed according to the type of the obstacle **Z**.

[0056] Therefore, the execution determination unit **332** determines that the execution condition of the offset travel is not satisfied when the distance **d1** with the division line is less than the lane reference distance **dref1**, when the distance **d2** with the curbstone **Y** is less than the step reference distance **dref2**, or when the distance **d3** with the obstacle **Z** is less than the obstacle reference distance **dref3**. On the other hand, when the distance **d1** with the division line is equal to or greater than the lane reference distance **dref1**, the distance **d2** with the curbstone **Y** is equal to or greater than the step reference distance **dref2**, and the distance **d3** with the obstacle **Z** is equal to or greater than the obstacle reference distance **dref3**, the execution determination unit **332** determines that the execution condition of the offset travel is satisfied.

[0057] The road condition of the vehicle **100** on the opposite side of the oncoming lane **Lo** may include elements other than the dividing lines, the step, and the obstacle. Therefore, the road condition on the side opposite to the opposite lane **Lo** side of the vehicle **100** may include, for example, whether another vehicle is traveling in the different traveling lane in a case where there is another traveling lane on the side opposite to the opposite lane **Lo** side of the traveling lane **Ld**, whether the other vehicle is approaching the vehicle **100** in a case where the other vehicle is traveling, and the like.

[0058] Specifically, the execution determination unit **332** performs image processing on the front image input from the outside camera **11** to recognize an object such as a demarcation line, a curbstone **Y**, and an obstacle **Z**. In addition, the execution determination unit **332** calculates the distance to the recognized object such as the curbstone **Y** and the obstacle **Z** based on the distance information input from the ranging sensor **12**. In particular, at this time, the execution determination unit **332** calculates not only the distance between the vehicle **100** and the curbstone **Y**, the obstacle **Z**, or the like, but also the distance between the vehicle **100** and the curbstone **Y**, the obstacle **Z**, or the like during traveling at the reference position.

[0059] The vehicle control unit **333** controls the vehicle **100**. In the present embodiment, the vehicle control unit **333** controls the operation of the vehicle **100** and controls the notification to the occupant. In particular, in the present embodiment, the vehicle control unit **333** can autonomously control the acceleration/deceleration and the steering of the vehicle **100** by autonomously controlling the operation of the vehicle actuator **21**. In addition, in the present embodiment, the vehicle control unit **333** controls notification to the occupant via HMI **15** information-providing device. The vehicle control unit **333** receives the front image input from the outside camera **11**, the distance information of the ranging sensor **12**, and the self-position information of the positioning sensor **13**. Further, an output value of the vehicle sensor **14** is input to the vehicle control unit **333**. Further, the vehicle control unit **333** outputs a control signal to the vehicle actuator **21** and outputs a notification signal to HMI **15**.

[0060] In the present embodiment, the vehicle control unit **333** controls the operation of the vehicle actuator **21** so that the vehicle **100** is autonomously traveled by the offset traveling when it is determined that the execution condition of the offset traveling is satisfied when the offset traveling is necessary. On the other hand, when it is determined that the execution condition of the offset travel is not satisfied when the offset travel is necessary, the vehicle control unit **333** performs at least one of restricting the autonomous driving function and notifying the occupant that the offset travel is impossible.

[0061] Specifically, in the present embodiment, the vehicle control unit **333** performs autonomous steering control of the vehicle **100** so as to perform the offset travel when the necessity determination unit **331** determines that the offset travel is necessary and the execution determination unit **332** determines that the execution condition of the offset travel is satisfied. That is, in such a case, the vehicle control unit **333** controls the vehicle actuator **21**, in particular, the

steering actuator, so as to perform the offset travel. Specifically, the vehicle control unit **333** calculates a traveling route of the vehicle **100** for performing offset traveling based on high-precision map information, self-position information, and the like, such as a front image input from the outside camera **11**. Then, the vehicle control unit **333** controls the steering actuator based on the output of the vehicle sensor **14** or the like so that the vehicle **100** travels along the travel route. [0062] Consequently, the vehicle **100** is autonomously steered so as to travel while maintaining a distance $D2$ between the center of the vehicle **100** and the end of the oncoming lane equal to or greater than a predetermined reference distance D_{ref} , as indicated by a solid line in FIG. **3**. In addition, in this case, the driver of the vehicle **100** does not need to control the steering of the vehicle **100**, and therefore, the vehicle control unit **333** does not cause HMI **15** information-providing device to issue a hands-on-request notification prompting the driver to grip the steering wheel.

[0063] On the other hand, in the present embodiment, the vehicle control unit **333** does not perform the autonomous steering control of the vehicle **100** when the necessity determination unit **331** determines that the offset travel is necessary and the execution determination unit **332** determines that the execution condition of the offset travel is not satisfied. That is, in such a case, the vehicle control unit **333** does not control the vehicle actuator **21**, in particular, the steering actuator. Therefore, in this case, the driver of the vehicle **100** needs to control the steering of the vehicle **100**. Therefore, the vehicle control unit **333** causes the information-providing device of HMI **15** to issue a hands-on request notification. Specifically, for example, the vehicle control unit **333** causes the display **16** to display a notification instructing the driver to grip the steering wheel, and causes the speaker **17** to output the notification.

[0064] As described above, in the present embodiment, in the case where the oncoming lane is close to the face-to-face traffic road, the autonomous steering control is basically performed so as to travel by the offset travel. Therefore, the oncoming vehicle is suppressed from traveling near the vehicle **100**, and the occupant's feeling of anxiety is reduced. However, in the present embodiment, when it is determined that the execution condition for executing the offset travel is not satisfied based on the road condition on the opposite side of the oncoming lane side of the vehicle **100**, the driver performs the steering control without performing the autonomous steering control. Thus, when the road condition on the side opposite to the oncoming lane side of the vehicle **100** is a condition in which, for example, a road shoulder or an obstacle is close, the autonomous steering control is not performed, and the steering control by the driver is performed. Therefore, in such a case, the occupant's feeling of anxiety is reduced.

Steering Process by Vehicle Control Device

[0065] FIG. **5** is a flowchart related to a steering process of the vehicle **100**. The illustrated steering process is executed by the processor **33** of ECU **30** at regular intervals.

[0066] When the steering process is started, first, the necessity determination unit **331** determines whether autonomous steering control is required (S11). The demand for the autonomous steering control is input by the occupant via the touch panel **18** or another input device on HMI **15** of the vehicle **100**, for example. Specifically, for example, ON.Math.OFF of the autonomous driving control including the autonomous steering control and the autonomous acceleration/deceleration control is selected by the occupant. Then, the necessity determination unit **331** determines that there is a request for the autonomous steering control when ON of the autonomous driving control is selected, and determines that there is no request for the autonomous steering control when OFF of the autonomous driving control is selected. When it is determined in S11 that the autonomous steering control is not required, the vehicle control unit **333** does not execute the autonomous steering control (S12). Therefore, the steering control by the driver is performed.

[0067] When it is determined in S11 that there is a request for autonomous steering control, the necessity determination unit **331** determines whether or not the distance D from the end of the oncoming lane is less than the reference distance D_{ref} , that is, whether or not offset travel is

required (S13). When it is determined in S13 that the distance D is equal to or larger than the reference distance Dref, hands-off is permitted, and therefore, the vehicle control unit 333 does not notify the hands-on request (S14). In addition, in such cases, the vehicle control unit 333 executes the autonomous steering control in the normal traveling without performing the off-set traveling (S15). Therefore, the vehicle control unit 333 controls the vehicle actuator 21, in particular, the steering actuator, so that the vehicle 100 travels in the center of the traveling lane.

[0068] When it is determined in S13 that the distance D is less than the reference distance Dref, that is, when it is determined that the offset travel is required, the execution determination unit 332 determines whether or not the execution condition of the offset travel is satisfied (S16). When it is determined in S16 that the condition for executing the offset-running is satisfied, the hands-off is allowed, and therefore, the vehicle control unit 333 does not perform the hands-on-request notification (S17). In addition, in such a case, the vehicle control unit 333 executes the autonomous steering control in the offset travel. Therefore, the vehicle control unit 333 controls the vehicle actuator 21, in particular, the steering actuator, so that the vehicle 100 travels on the side opposite to the opposite lane side from the center of the traveling lane. On the other hand, when it is determined that S16 does not satisfy the condition for executing the offset-running, the vehicle control unit 333 causes HMI 15 information-providing device to issue a hands-on-request notification (S19). In addition, in such cases, the vehicle control unit 333 does not execute the autonomous steering control after it is detected by the driver that the steering wheel has been gripped (S20). Therefore, the steering control by the driver is performed.

Modification

[0069] Hereinafter, a modification of the above-described embodiment will be described. In the above-described embodiment, when the offset traveling is required and the execution condition of the offset traveling is satisfied, the autonomous steering control by the offset traveling is performed without the hands-on request notification. On the other hand, when the offset traveling is necessary and the execution condition of the offset traveling is not satisfied, a hands-on request notification is performed, and the autonomous steering control is not performed.

[0070] On the other hand, in the first modification, when the offset travel is necessary and the execution condition of the offset travel is satisfied, the vehicle control unit 333 performs the autonomous steering control by the offset travel. Therefore, in this case, the vehicle control unit 333 does not perform the hands-on request notification. At this time, the vehicle control unit 333 controls the speed of the vehicle 100 so that the speed of the vehicle 100 becomes the same as the speed in the normal travel in which the offset travel is not required. That is, the vehicle control unit 333 controls the vehicle actuator 21, in particular, the drive actuator and the braking actuator so that the speed of the vehicle 100 becomes such a speed.

[0071] On the other hand, in the first modification, when the offset travel is required and the execution condition of the offset travel is not satisfied, the vehicle control unit 333 executes the autonomous steering control in the normal travel without the offset travel. In this case, the vehicle control unit 333 does not issue a hands-on request notification. In addition, the vehicle control unit 333 controls the speed of the vehicle 100 so that the speed of the vehicle 100 becomes a speed lower than the speed in the normal travel in which the offset travel is not necessary. That is, when it is determined that the execution condition of the offset travel is not satisfied, the vehicle control unit 333 decreases the speed of the vehicle 100 as compared with the case where it is determined that the execution condition of the offset travel is satisfied.

[0072] At this time, the vehicle control unit 333 may control the speed of the vehicle 100 such that the speed is slower by a predetermined constant speed or a constant ratio than the speed in normal traveling. Alternatively, the vehicle control unit 333 may control the speed of the vehicle 100 such that the speed is slower by a speed corresponding to the distance D from the end of the oncoming lane than the speed in normal traveling. In this case, the speed of the vehicle 100 is controlled so that the speed of the vehicle 100 becomes slower as the distance D from the end of the oncoming

lane becomes shorter.

[0073] FIG. 6 is a flowchart related to a steering process of the vehicle 100 according to the first modification. The illustrated steering process is executed by the processor 33 of ECU 30 at regular intervals. S31-S35 in FIG. 6 are the same as those in S11-S13, S15, S16 of FIG. 5, and therefore will not be described.

[0074] When it is determined in S35 that the execution condition of the offset travel is satisfied, the vehicle control unit 333 executes the autonomous steering control in the offset travel (S36). At this time, the vehicle control unit 333 controls the speed of the vehicle 100 so that the speed becomes the same as the speed in the normal travel in which the offset travel is not performed. On the other hand, when it is determined in S35 that the execution condition of the offset travel is not satisfied, the vehicle control unit 333 executes the autonomous steering control in the normal travel without performing the offset travel. At this time, the vehicle control unit 333 limits the speed of the vehicle 100 to a speed slower than the speed in the normal travel in which the offset travel is not necessary.

[0075] In the first modification, when it is determined that the execution condition for executing the offset travel based on the road condition on the opposite side to the oncoming lane side of the vehicle 100 is not satisfied, the autonomous steering control is performed, but the speed of the vehicle 100 is reduced. This reduces the feeling of anxiety for the occupant.

[0076] In addition, in the second modification, when the offset travel is necessary and the execution condition of the offset travel is satisfied, the vehicle control unit 333 performs autonomous steering control by the offset travel. Further, in the second modification, when the offset travel is necessary and the execution condition of the offset travel is satisfied, the vehicle control unit 333 executes the autonomous steering control in the normal travel without performing the offset travel. In addition, in this case, the vehicle control unit 333 issues a notification indicating a warning to the occupant of the vehicle 100. Specifically, the vehicle control unit 333 causes HMI 15 information-providing device to issue a notification that warns that the vehicle 100 is traveling in the vicinity of the oncoming lane. Thus, the driver of the vehicle 100 can determine whether it is necessary to perform the steering control of the vehicle 100 by himself/herself, and can perform the steering control as necessary. Therefore, it is suppressed that the oncoming vehicle suddenly travels near the vehicle 100 without any warning, and thus the occupant is less concerned.

[0077] While preferred embodiments according to the present disclosure have been described above, the present disclosure is not limited to these embodiments, and various modifications and changes can be made within the scope of the claims.

Claims

1. A vehicle control device that is able to autonomously control steering of a vehicle, the vehicle control device comprising: a necessity determination unit that determines necessity of offset travel in which the vehicle is offset from a center of a traveling lane, in which the vehicle is traveling, to a side opposite to an opposite lane, in which a vehicle travels in a direction opposite to the traveling lane, based on a situation of the opposite lane; an execution determination unit that determines whether an execution condition for executing the offset travel is met based on a road condition on a side of the vehicle opposite to the opposite lane when the offset travel is necessary; and a vehicle control unit that controls the vehicle, wherein: the vehicle control unit causes the vehicle to travel using the offset travel when it is determined that the execution condition is met when the offset travel is necessary; and the vehicle control unit performs at least one of restricting an autonomous driving function and notifying an occupant when it is determined that the execution condition is not met when the offset travel is necessary.

2. The vehicle control device according to claim 1, wherein the vehicle control unit is configured to: make a hands-on request notification that prompts a driver of the vehicle to grip a steering wheel when it is determined that the execution condition is not met when the offset travel is

necessary; and make no hands-on request notification when it is determined that the execution condition is met when the offset travel is necessary.

3. The vehicle control device according to claim 1, wherein the vehicle control unit reduces a speed of the vehicle when it is determined that the execution condition is not met when the offset travel is necessary, as compared with when it is determined that the execution condition is met when the offset travel is necessary.

4. The vehicle control device according to claim 1, wherein the vehicle control unit causes the vehicle to travel without using the offset travel and notifies an occupant of the vehicle for warning when it is determined that the execution condition is not met when the offset travel is necessary.

5. A vehicle control method that is able to autonomously control steering of a vehicle, the vehicle control method comprising: determining necessity of offset travel in which the vehicle is offset from a center of a traveling lane, in which the vehicle is traveling, to a side opposite to an opposite lane, with traffic in a direction opposite to the traveling lane, based on a situation of the opposite lane; determining whether an execution condition for executing the offset travel is met based on a road condition on a side of the vehicle opposite to the opposite lane when the offset travel is necessary; causing the vehicle to travel using the offset travel when it is determined that the execution condition is met when the offset travel is necessary; and performing at least one of restricting an autonomous driving function and notifying an occupant when it is determined that the execution condition is not met when the offset travel is necessary.
